

# Phenotyping Algorithm for Autoimmune Disease (AID) V4

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Updates (Compared to V3):

- Updated serology tests LOINC codes

Updates (Compared to V2):

- Combined coding dictionary for cases and controls files
- Added data dictionary for co-variables extraction

## I. Background and Significance

The goal of the AID phenotype definition is to identify an individual as case or control using structured electronic health record (EHR) elements. The autoimmune disease case phenotype spans 6 medical specialties, 9 groups of disease, and 51 diseases (46 non-overlapping diseases) (Table 1 and 2). Groups of diseases are loosely based on organ system. The 5 diseases causing the discrepancy between the 51 diseases and 46 non-overlapping diseases are due to the diseases belonging to parent diseases that are included in the definition (i.e., Giant Cell Arteritis and Cerebral Arteritis are children concepts to the parent concept Arteritis, Optic Papillitis is a child concept to Optic Neuritis, Ocular Cicatricial Pemphigoid is a child concept to parent Pemphigoid, Dermatomyositis is a child concept of Polymyositis). In addition to excluding autoimmune diseases, the autoimmune disease control phenotype excludes auto-inflammatory diseases and the presence of 10 sets of antibody tests (Table 3).

Each disease is defined by a collection of ICD9CM, ICD10CM, and/or SNOMED codes. A total of 1528 diagnosis codes were used to determine case status, and 49960 diagnosis codes and measurement codes were used for controls.

## II. Development

- Autoimmune disease case and control phenotype was developed using structured EHR data using domain knowledge and standards.
- Validation was performed against expert opinion, including local experts from Rheumatology, Nephrology, Dermatology, Neurology, and Gastroenterology.

## III. Definition

A flow diagram depicting how participants in a cohort are sorted into case and control cohort is in Figure 1. Pseudocode is described in Figure 2.

### III.i. Autoimmune Disease Case

The presence of at least one autoimmune disease qualifies the patient to be a case.

A patient meeting the following condition (Case Cohort: Condition A) is defined as having at least one autoimmune disease:

Case Cohort: Condition A. Any 3 diagnosis codes on distinct days within the same disease with the first and last diagnoses at least 7 days apart. Codes defining each disease are listed in AIDalgorithm\_V1\_coding\_cases.csv and are grouped by disease name. For patients with diagnosis codes for type 1 diabetes mellitus (T1DM), an additional exclusion criterion of no type 2 diabetes mellitus (T2DM) codes is applied. Codes defining T2DM are listed in file AIDalgorithm\_V1\_coding\_cases.csv under variable name, "Type 2 Diabetes Mellitus (T2DM)". Patients meeting the case eligibility criteria will be tagged as belonging to group(s) and disease(s) based on the presence of diagnosis code(s).

### III.ii. Autoimmune Disease Control

The absence of any selected auto-inflammatory condition, autoimmune disease, and positive serology test qualifies a patient to be a control.

A patient not having any auto-inflammatory condition, autoimmune disease, and positive serology test must meet both of the following conditions (Control Cohort: Condition A and Control Cohort: Condition B):

Control Cohort: Condition A. No autoimmune and auto-inflammatory diagnosis codes. Codes are listed in file AIDalgorithm\_V1\_coding\_control.csv under variable name, "Auto-inflammatory + Autoimmune (SNOMED)".

Control Cohort: Condition B. No instances of any positive serologies as defined by institutional and assay recommendations from a list of serologies (Table 3). Codes are listed in file AIDalgorithm\_V1\_coding\_control.csv under variable names starting with "Serology".

Note: Opening \*.csv in Excel strips the ".0" in ICD9 codes.

## IV. Autoimmune Disease Implementation

The coding dictionary for the algorithm variables (i.e., autoimmune diseases, auto-inflammatory diseases, serologies) is index by condition in Table 4. Table 4 refers to 2 csv files: AIDalgorithm\_V1\_coding\_cases.csv and AIDalgorithm\_V1\_coding\_controls.csv.

The coding dictionary is also located on OHDSI ATLAS

(<http://www.ohdsi.org/web/atlas/#/cohortdefinitions>) under Cohorts ID 102998 for cases and Cohorts ID 919472 for controls.

### IV.I Coding dictionary CSV files

Each disease, or subphenotype, is defined by one or more codes belonging to one or more terminology vocabulary. For example, the codes for the subphenotype, “Vasculitis: Takayasu's disease”, are located in AIDalgorithm\_V1\_coding\_cases.csv and are the following:

|                          |         |           |                                 |
|--------------------------|---------|-----------|---------------------------------|
| <input type="checkbox"/> | ICD10CM | M31.4     | Aortic arch syndrome [Takayasu] |
| <input type="checkbox"/> | SNOMED  | 359789008 | Takayasu's disease              |
| <input type="checkbox"/> | ICD9CM  | 446.7     | Takayasu's disease              |

### IV.II Using OHDSI ATLAS

1. Go to <http://www.ohdsi.org/web/atlas/#/cohortdefinitions>
2. Click on “Cohorts” on the left panel
3. Search for case or controls by typing 102998 or 919472, respectively, into “Filter:”
4. Click on the cohort in the search results
5. Select the “Concept Set” Tab located on the right panel under title
6. To export concept sets, click “Export All Concept Sets to CSV”

## V. Covariates

Demographic information: eMERGE\_ID, Age, Sex, Race, Ethnicity, Case\_Control\_Unknown\_Status.

AID subphenotype: eMERGE\_ID, subPhenotype.

AID algorithm relevant diagnosis: eMERGE\_ID, Age\_diagnosis, diagnosis\_code, code\_type.

Serology test (Table 3): eMERGE\_ID, serology\_test\_name, age\_at\_serology\_test, serology\_test\_result.

Complement (C3 and C4) test: eMERGE\_ID, complement\_test\_name, age\_at\_complement\_test, complement\_test\_result

Quantitative immunoglobulins (IgA, IgG, and IgM) test: eMERGE\_ID, immunoglobulin\_test\_name, age\_at\_immunoglobulin\_test, immunoglobulin\_test\_result

## VI. Accompanying Documents

The documents for the Autoimmune Disease Phenotype are as follows:

1. AutoimmuneDiseaseAlgorithm\_V4.pdf – Main document containing the AID algorithm definition.
2. Validation Instructions.docx – Validation instructions for secondary site.
3. AIDalgorithm\_coding\_V3.txt – Lists of codes used for Autoimmune Disease Case and Control Phenotype (corresponds to Table 4). This includes mapped OMOP vocabulary (conceptId).
4. AIDalgorithm\_dd\_V1.xls – AID co-variable data dictionary.

## Figures and Tables

Table 1. The autoimmune disease case phenotype spans 9 groups of disease, and 51 diseases (46 non-overlapping diseases) or subphenotypes

| D  | Grouping Name | Disease Name                                |
|----|---------------|---------------------------------------------|
| 1  | Arthritis     | Ankylosing spondylitis                      |
| 2  | Arthritis     | Bechet's disease                            |
| 3  | Arthritis     | Palindromic rheumatism                      |
| 4  | Arthritis     | Polymyalgia rheumatica                      |
| 5  | Arthritis     | Psoriatic arthritis                         |
| 6  | Arthritis     | Reiter's syndrome                           |
| 7  | Arthritis     | Rheumatoid arthritis (RA)                   |
| 8  | Connective    | Lupus erythematosus                         |
| 9  | Connective    | Mixed Connective Tissue Disease (MCTD)      |
| 10 | Connective    | Sarcoidosis                                 |
| 11 | Connective    | Scleroderma                                 |
| 12 | Connective    | Sjogren's syndrome                          |
| 13 | Endocrine     | Graves' Disease                             |
| 14 | Endocrine     | Hashimoto's thyroiditis                     |
| 15 | Endocrine     | T1D                                         |
| 16 | GI            | Autoimmune hepatitis                        |
| 17 | GI            | Celiac Disease                              |
| 18 | GI            | Crohn's disease                             |
| 19 | GI            | Primary biliary cholangitis (PBC)           |
| 20 | GI            | Ulcerative colitis (UC)                     |
| 21 | Heme          | Antiphospholipid syndrome (APS)             |
| 22 | Heme          | Autoimmune hemolytic anemia (AIHA)          |
| 23 | Heme          | Autoimmune neutropenia                      |
| 24 | Heme          | Evans syndrome                              |
| 25 | Heme          | Thrombocytopenic purpura (TTP)              |
| 26 | Muscle        | Dermatomyositis* <sup>1</sup>               |
| 27 | Muscle        | Polymyositis* <sup>1</sup>                  |
| 28 | Muscle        | Inflammatory and immune myopathies          |
| 29 | Neuro         | Guillain-Barre Syndrome                     |
| 30 | Neuro         | Lambert-Eaton syndrome                      |
| 31 | Neuro         | Multiple sclerosis                          |
| 32 | Neuro         | Myasthenia gravis                           |
| 33 | Neuro         | Myelitis transversa                         |
| 34 | Neuro         | Optic neuritis* <sup>2</sup>                |
| 35 | Neuro         | Optic Papillitis* <sup>2</sup>              |
| 36 | Neuro         | Schilder's disease                          |
| 37 | Skin          | Alopecia areata                             |
| 38 | Skin          | Dermatitis herpetiformis                    |
| 39 | Skin          | Pemphigoid* <sup>3</sup>                    |
| 40 | Skin          | Ocular cicatricial pemphigoid* <sup>3</sup> |
| 41 | Skin          | Pemphigus                                   |
| 42 | Skin          | Psoriasis                                   |
| 43 | Skin          | Pyoderma                                    |
| 44 | Skin          | Raynaud                                     |
| 45 | Skin          | Vitiligo                                    |
| 46 | Vasculitis    | Arteritis* <sup>4</sup>                     |
| 47 | Vasculitis    | Cerebral Arteritis* <sup>4</sup>            |
| 48 | Vasculitis    | Giant Cell Arteritis* <sup>4</sup>          |
| 49 | Vasculitis    | Goodpasture's syndrome                      |
| 50 | Vasculitis    | Granulomatosis                              |
| 51 | Vasculitis    | Takayasu's disease                          |

\* Denotes diseases that overlap with other diseases and are combined into one sub-phenotype.

Table 2. Medical Specialty and Groupings

| Medical Specialty | Name of Group of Diseases |
|-------------------|---------------------------|
| Rheumatology      | Arthritis                 |
|                   | Connective Tissue         |
|                   | Muscular (Muscle)         |
|                   | Vasculitis                |
| Endocrinology     | Endocrine                 |
| Gastroenterology  | Gastrointestinal (GI)     |
| Hematology        | Hematologic (Heme)        |
| Neurology         | Neurological (Neuro)      |
| Dermatology       | Skin                      |

Table 3. Serology Antibody Tests

| #  | Antibody Tests:                                            |
|----|------------------------------------------------------------|
| 1  | Anti Nuclear Antibody (ANA)                                |
| 2  | Anti-Cytoplasmic Neutrophil Antibodies (ANCA)*             |
| 3  | Anti-DNA Antibody (dsDNA)                                  |
| 4  | Cyclic Citrullinated Peptide Antibody (CCP)                |
| 5  | Rheumatoid Factor (RF)                                     |
| 6  | Beta 2 Glycoprotein I Antibody (B2 Glycoprotein 1, B2Glyc) |
| 7  | RNA Polymerase 3 Antibody (RNA PIII, RNAP3)                |
| 8  | Anti-Cardiolipin Antibodies (Cardiolipin)*                 |
| 9  | Centromere Antibody IgG (Centromere)                       |
| 10 | Extractable Nuclear Antibodies (ENA)*                      |

|                                                             |
|-------------------------------------------------------------|
| Anti-Cytoplasmic Neutrophil Antibodies include:             |
| Serum Anti Neutrophil Cytoplasmic Antibody C Test (cANCA)   |
| Serum Anti Neutrophil Cytoplasmic Antibody P Test (pANCA)   |
|                                                             |
| Anti-Cardiolipin Antibodies include:                        |
| Anti-Cardiolipin IgG Antibody Measurement (Cardiolipin_IgG) |
| Anti-Cardiolipin IgM Antibody Measurement (Cardiolipin_IgM) |
|                                                             |
| Extractable Nuclear Antibodies include:                     |
| Jo 1 Autoantibody (J01)                                     |
| Ribonucleoprotein (U1RNP) Autoantibody (antiRNP)            |
| Scleroma Autoantibody(Scl 70)                               |
| Smith Autoantibodies (Sm)                                   |
| Sjogren's Syndrome A (SS-A) Antibodies IgG (SSA)            |
| Sjogren's Syndrome B (SS-B) Antibodies IgG (SSB)            |

Table 4. An index to the code dictionary sorted by each condition and description of data

| ID | Condition, Description of Data                                                  | Filename, Variable Names within file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1 | Case Cohort: Condition A, Autoimmune diseases                                   | <div><div>AIDalgorithm_V1_coding_cases.csv,</div><div>Arthritis: Ankylosing spondylitis</div><div>Arthritis: Behcet’s disease</div><div>Arthritis: Palindromic rheumatism</div><div>Arthritis: Polymyalgia rheumatica</div><div>Arthritis: Psoriatic arthritis</div><div>Arthritis: Reiter’s syndrome</div><div>Arthritis: Rheumatoid arthritis (RA)</div><div>Connective: Lupus erythematosus</div><div>Connective: Mixed Connective Tissue Disease (MCTD)</div><div>Connective: Sarcoidosis</div><div>Connective: Scleroderma</div><div>Connective: Sjogren’s syndrome</div><div>Endocrine: Graves' Disease</div><div>Endocrine: Hashimoto’s thyroiditis</div><div>Endocrine: T1D</div><div>GI: Autoimmune hepatitis</div><div>GI: Celiac Disease</div><div>GI: Crohn’s disease</div><div>GI: Primary biliary cholangitis (PBC)</div><div>GI: Ulcerative colitis (UC)</div><div>Heme: Antiphospholipid syndrome (APS)</div><div>Heme: Autoimmune hemolytic anemia (AIHA)</div><div>Heme: Autoimmune neutropenia</div><div>Heme: Evans syndrome</div><div>Heme: Thrombocytopenic purpura (TTP)</div><div>Muscle: Dermatomyositis</div><div>Muscle: Inflammatory and immune myopathies</div><div>Muscle: Polymyositis</div><div>Neuro: Guillain-Barre Syndrome</div><div>Neuro: Lambert-Eaton syndrome</div><div>Neuro: Multiple sclerosis</div><div>Neuro: Myasthenia gravis</div><div>Neuro: Myelitis transversa</div><div>Neuro: Optic neuritis</div><div>Neuro: Optic Papillitis</div><div>Neuro: Schilder's disease</div><div>Skin: Alopecia areata</div><div>Skin: Dermatitis herpetiformis</div><div>Skin: Ocular cicatricial pemphigoid</div><div>Skin: Pemphigoid</div><div>Skin: Pemphigus</div><div>Skin: Psoriasis</div><div>Skin: Pyoderma</div><div>Skin: Raynaud</div><div>Skin: Vitiligo</div><div>Vasculitis: Arteritis</div><div>Vasculitis: Cerebral Arteritis</div><div>Vasculitis: Giant Cell Arteritis</div><div>Vasculitis: Goodpasture's syndrome</div><div>Vasculitis: Granulomatosis</div><div>Vasculitis: Takayasu's disease</div></div> |
| A2 | Case Cohort: Condition A, T2DM                                                  | <div><div>AIDalgorithm_V1_coding_cases.csv,</div><div>Type 2 Diabetes Mellitus (T2DM)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| A3 | Control Cohort: Condition A, Autoimmune diseases and Auto-inflammatory diseases | <div><div>AIDalgorithm_V1_coding_controls.csv,</div><div>Auto-inflammatory + Autoimmune</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| B1 | Control Cohort: Condition B, Serologies                                         | <div><div>AIDalgorithm_V1_coding_controls.csv,</div><div>Serology: ANA</div><div>Serology: ANCA</div><div>Serology: dsDNA</div><div>Serology: CCP</div><div>Serology: RF</div><div>Serology: B2 Glycoprotein 1</div><div>Serology: RNA PIII</div><div>Serology: Cardiolipin</div><div>Serology: Centromere IgG</div><div>Serology: ENA</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |